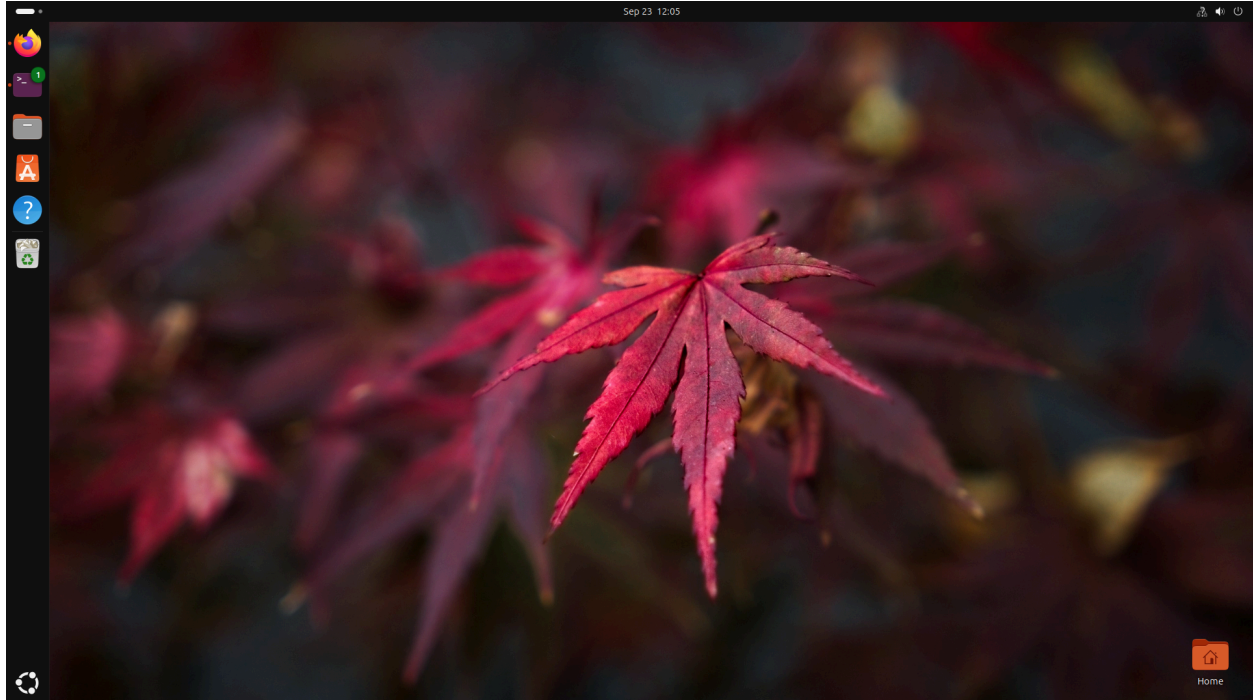


Exercise 1 Learning Linux Desktop Features Answers

1.

1. In start menu we have Calculator, Firmware updater , language , Clock, Resources, Security center , Software updater, Settings,Terminal,Text editor,Characters, utilities,System.

Desktop background changed



2.

3.Screenshot

4.list of the information in Network settings

Details

- Link speed- 1000mbps
- IPv4 address-192.168.2.121
- IPv6 address- fe80::fe4c:eaff:fe23:9cfb
- Hardware address- FC:4C:EA:23:9C:FB
- Default route- 192.168.1.254
- DNS- 192.168.1.254 192.168.1.11

5.Apps/applications

- Advanced Network configuration:Manages complex network setups like VPNs, bridges, and static IPs.
- App center :The built-in software store used to find, install, and remove applications.
- Calculator :A standard tool for basic and advanced scientific calculations.
- Characters : A utility to find and insert unique symbols, emojis, and punctuation marks.
- Clocks : Manages world clocks, alarms, stopwatches, and timers.
- Disk : A utility to format, partition, benchmark, and inspect hard drives or SSDs.
- Disk usage analyzer : Visually maps out storage space to show which files or folders take up the most room.
- Document viewer : Opens PDFs, Comic Books, and other document formats.

- Files : The default file manager used to browse your folders and documents.
- Firefox : The default, privacy-focused open-source web browser.
- Firmware updater : Installs hardware-level updates (like BIOS or device firmware).
- Fonts : Previews, manages, and installs new typefaces on your system.
- Help : The offline user documentation guide for the operating system.
- Image Viewer : A lightweight utility to view and rotate pictures.
- Language support : Installs and configures system languages, keyboard layouts, and regional formats.
- Logs : A diagnostic tool to view system events and error reports for troubleshooting.
- Passwords and Keys : Manages encryption keys and saved passwords via the GNOME Keyring.
- Resources : Monitors CPU, memory, network, and GPU usage in real-time.
- Security center : A modern dashboard to monitor firewall status, device security, and permissions.
- Settings : The central hub for customizing your system (display, sound, bluetooth, backgrounds).
- Software updater: Fetches and installs updates for your operating system and installed apps.
- Sysproof : A specialized diagnostic or profiling utility to verify system integrity and performance.
- Terminal : The command-line interface (CLI) used to interact directly with the operating system using text commands.
- Text editor : A simple, clean app for writing notes or editing code (typically Gedit or GNOME Text Editor).

2.Github username: rohitkumbar2003

3. Linux

- IBAB Workshop-Comp15.ibab.ac.in :The network name of the computer.
- 7.0.0-14-generic #14-Ubuntu :The specific version or build number of the current operating system kernel.
- SMP PREEMPT_DYNAMIC :Stands for **Symmetric Multi-Processing**, indicating that the kernel supports multiple processors or CPU cores.
- Mon Apr 13 11:09:53 :The exact date and time the current kernel was compiled and built.
- UTC 2026 x86_64 : The hardware architecture
- GNU/Linux :The specific hardware platform type

```
ibab@IBAB-Workshop-Comp15:~$ git config --global user.mail rohitkumbar2003@gmail.com
ibab@IBAB-Workshop-Comp15:~$ uname -a
Linux IBAB-Workshop-Comp15.ibab.ac.in 7.0.0-14-generic #14-Ubuntu SMP PREEMPT_DYNAMIC Mon Apr 13 11:09:53 UTC 2026 x86_64 GNU/Linux
ibab@IBAB-Workshop-Comp15:~$
```

uname -r : 7.0.0-14-generic	build_number
uname -n: IBAB Workshop-Comp15.ibab.ac.in	host name
uname -m: x86_64	hardware architecture
uname -s: Linux	Operating system
uname -o: GNU/Linux	hardware platform

```
ibab@IBAB-Workshop-Comp15:~$ uname -r
7.0.0-14-generic
ibab@IBAB-Workshop-Comp15:~$ uname -n
IBAB-Workshop-Comp15.ibab.ac.in
ibab@IBAB-Workshop-Comp15:~$ uname -m
x86_64
ibab@IBAB-Workshop-Comp15:~$ uname -s
Linux
ibab@IBAB-Workshop-Comp15:~$ uname -o
GNU/Linux
```

4.

Pwd : previous working directory /home/ibab

lbab

Hostname is displayed as IBAB Workshop-Comp15.ibab.ac.in and it is also present in uname -a in 2nd line

/home/ibab path of home directory

Ls and ls -a in home directory

```
ibab@IBAB-Workshop-Comp15:~$ ls
Desktop Documents Downloads Music Pictures Public R Templates Videos apt.conf snap
ibab@IBAB-Workshop-Comp15:~$ ls -a
. . . TurboFloat .bash_history .bash_logout .bashrc .cache .config .gitconfig .local .prof .profile .rstudio-desktop Desktop Documents Downloads Music Pictures Public R Templates Videos apt.conf snap
```

```
ibab@IBAB-Workshop-Comp15:~$ cd
ibab@IBAB-Workshop-Comp15:~$ pwd
/home/ibab
ibab@IBAB-Workshop-Comp15:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos apt.conf snap
ibab@IBAB-Workshop-Comp15:~$ -a
-a: command not found
ibab@IBAB-Workshop-Comp15:~$ la -a
. . . .bash_history .bash_logout .bashrc .cache .config .gitconfig .local .profile Desktop Documents Downloads Music Pictures Public Templates Videos apt.conf snap
ibab@IBAB-Workshop-Comp15:~$ cd /
ibab@IBAB-Workshop-Comp15:/ $ cd ~
ibab@IBAB-Workshop-Comp15:~$ cd
ibab@IBAB-Workshop-Comp15:~$ cd..
cd..: command not found
ibab@IBAB-Workshop-Comp15:~$ cd ..
ibab@IBAB-Workshop-Comp15:/home$ cd .
ibab@IBAB-Workshop-Comp15:/home$ cd -
/home
```

cd commands

6.-R : its gives information of all sub directories

-r : it provides files in reverse order

7.By command pwd

8.Command : ls -lrt

```

ibab@IBAB-Workshop-Comp15:~$ ls -lrt
total 80
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Desktop
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Templates
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Music
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Public
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Videos
-rw-rw-r-- 1 ibab ibab 279 Sep 17 14:25 apt.conf
drwxr-xr-x 3 ibab ibab 4096 Sep 17 14:38 Pictures
drwx----- 6 ibab ibab 4096 Sep 17 15:00 snap
drwxrwxr-x 3 ibab ibab 4096 Sep 17 15:25 R
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 lab
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 2
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 lab2
drwxr-xr-x 5 ibab ibab 4096 Sep 18 10:43 Documents
-rw-rw-r-- 3 ibab ibab 771 Sep 18 12:17 testdir-hardlink
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:26 file2
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:30 file3
drwxr-xr-x 5 ibab ibab 4096 Sep 18 12:30 Docs
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:31 file4
lrwxrwxrwx 1 ibab ibab 5 Sep 18 12:46 testdir-symlink -> file1
-rw-rw-r-- 1 ibab ibab 11 Sep 18 13:01 file1_hardlink
drwxr-xr-x 2 ibab ibab 4096 Sep 18 14:04 Downloads

```

9. Using `ls -lrt` we can get detailed information of files with modification time and in sorted order where old files are pushed on top making appear on first , but by `ls -l` we can get only long listing of files.

```

ibab@IBAB-Workshop-Comp15:~$ ls -lrt
total 80
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Desktop
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Templates
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Music
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Public
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Videos
-rw-rw-r-- 1 ibab ibab 279 Sep 17 14:25 apt.conf
drwxr-xr-x 3 ibab ibab 4096 Sep 17 14:38 Pictures
drwx----- 6 ibab ibab 4096 Sep 17 15:00 snap
drwxrwxr-x 3 ibab ibab 4096 Sep 17 15:25 R
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 lab
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 2
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 lab2
drwxr-xr-x 5 ibab ibab 4096 Sep 18 10:43 Documents
-rw-rw-r-- 3 ibab ibab 771 Sep 18 12:17 testdir-hardlink
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:26 file2
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:30 file3
drwxr-xr-x 5 ibab ibab 4096 Sep 18 12:30 Docs
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:31 file4
lrwxrwxrwx 1 ibab ibab 5 Sep 18 12:46 testdir-symlink -> file1
-rw-rw-r-- 1 ibab ibab 11 Sep 18 13:01 file1_hardlink
drwxr-xr-x 2 ibab ibab 4096 Sep 18 14:04 Downloads
ibab@IBAB-Workshop-Comp15:~$ ls -l
total 80
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 2
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Desktop
drwxr-xr-x 5 ibab ibab 4096 Sep 18 12:30 Docs
drwxr-xr-x 5 ibab ibab 4096 Sep 18 10:43 Documents
drwxr-xr-x 2 ibab ibab 4096 Sep 18 14:04 Downloads
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Music
drwxr-xr-x 3 ibab ibab 4096 Sep 17 14:38 Pictures
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Public
drwxrwxr-x 3 ibab ibab 4096 Sep 17 15:25 R
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Templates
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Videos
-rw-rw-r-- 1 ibab ibab 279 Sep 17 14:25 apt.conf
-rw-rw-r-- 1 ibab ibab 11 Sep 18 13:01 file1_hardlink
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:26 file2
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:30 file3
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:31 file4
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 lab
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 lab2
drwx----- 6 ibab ibab 4096 Sep 17 15:00 snap
-rw-rw-r-- 3 ibab ibab 771 Sep 18 12:17 testdir-hardlink
lrwxrwxrwx 1 ibab ibab 5 Sep 18 12:46 testdir-symlink -> file1

```

10.

ls -aIF : Includes all hidden files and current directory, working directory too and display all the information of the directories and files

ls -A : Only directory and file names are displayed which Includes all directories including hidden files , no current and parent directory

ls -CF : Only directory and files names are displayed, Do not include hidden and parent directories

```

drwx----- 6 ibab ibab 4096 Sep 17 15:00 snap
-rw-rw-r-- 3 ibab ibab 771 Sep 18 12:17 testdir-hardlink
lrwxrwxrwx 1 ibab ibab 5 Sep 18 12:46 testdir-symlink -> file1
ibab@IBAB-Workshop-Comp15:~$ ls -aLF
total 144
drwxr-xr-x 22 ibab ibab 4096 Sep 18 16:58 ./
drwxr-xr-x 6 root root 4096 Sep 18 11:51 ../
-rw-rw-r-- 1 ibab ibab 399 Sep 18 16:58 .Rdata
-rw-r--r-- 1 ibab ibab 1663 Sep 18 16:58 .Rhistory
drwxrwxr-x 2 ibab ibab 4096 Sep 17 17:27 TurboFloat/
-rw-r--r-- 1 ibab ibab 4377 Sep 18 14:02 .bash_history
-rw-r--r-- 1 ibab ibab 220 Sep 18 11:51 .bash_logout
-rw-r--r-- 1 ibab ibab 3793 Sep 18 10:58 .bashrc
drwxr-xr-x 17 ibab ibab 4096 Sep 18 08:31 .cache/
drwx----- 17 ibab ibab 4096 Sep 17 17:27 .config/
-rw-rw-r-- 1 ibab ibab 65 Sep 17 14:37 .gitconfig
drwx----- 4 ibab ibab 4096 Sep 17 14:07 .local/
drwxrwxr-x 2 ibab ibab 4096 Sep 17 17:27 .prof/
-rw-r--r-- 1 ibab ibab 807 Sep 18 11:51 .profile
drwxrwxr-x 2 ibab ibab 4096 Sep 17 17:27 .rstudio-desktop/
drwxrwxr-x 2 ibab ibab 4096 Sep 18 16:35 2/
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Desktop/
drwxr-xr-x 5 ibab ibab 4096 Sep 18 12:30 Docs/
drwxr-xr-x 5 ibab ibab 4096 Sep 18 10:43 Documents/
drwxr-xr-x 2 ibab ibab 4096 Sep 18 14:04 Downloads/
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Music/
drwxr-xr-x 3 ibab ibab 4096 Sep 17 14:38 Pictures/
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Public/
drwxrwxr-x 3 ibab ibab 4096 Sep 17 15:25 R/
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Templates/
drwxr-xr-x 2 ibab ibab 4096 Sep 17 14:07 Videos/
-rw-rw-r-- 1 ibab ibab 279 Sep 17 14:25 apt.conf
-rw-rw-r-- 1 ibab ibab 11 Sep 18 13:01 file1_hardlink
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:26 file2
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:30 file3
-rw-rw-r-- 1 ibab ibab 12 Sep 18 12:31 file4
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 lab/
drwxrwxr-x 2 ibab ibab 4096 Sep 18 10:35 lab2/
drwx----- 6 ibab ibab 4096 Sep 17 15:00 snap/
-rw-rw-r-- 3 ibab ibab 771 Sep 18 12:17 testdir-hardlink
lrwxrwxrwx 1 ibab ibab 5 Sep 18 12:46 testdir-symlink -> file1
ibab@IBAB-Workshop-Comp15:~$ ls -A
.Rdata TurboFloat .bash_logout .cache .gitconfig .prof .rstudio-desktop Desktop Documents Music Public Templates apt.conf file2 file4 lab2 testdir-hardlink
.Rhistory .bash_history .bashrc .config .local .profile 2
ibab@IBAB-Workshop-Comp15:~$ ls -CF
2/ Desktop/ Docs/ Documents/ Downloads/ Music/ Pictures/ Public/ R/ Templates/ Videos/ apt.conf file1_hardlink file2 file3 file4 lab/ lab2/ snap/ testdir-hardlink testdir-symlink@

```

11.

```

ibab@IBAB-Workshop-Comp15:~/sails_python_labs$ mkdir -p ~/lab1/Q10/q1
ibab@IBAB-Workshop-Comp15:~/sails_python_labs$ mkdir -p ~/lab1/Q10/q2
ibab@IBAB-Workshop-Comp15:~/sails_python_labs$ cd lab1
ibab@IBAB-Workshop-Comp15:~/lab1$ cd Q10
ibab@IBAB-Workshop-Comp15:~/lab1/Q10$ cd q1
ibab@IBAB-Workshop-Comp15:~/lab1/Q10/q1$ cp -r ~/lab1/Q10/q1 ~/lab1/Q10/q1_copy
ibab@IBAB-Workshop-Comp15:~/lab1/Q10/q1$ rm -ri ~/lab1/Q10/q1_copy

```